

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Steroids induce remission in IBD — can they be used long-term for maintenance?
2. Which enzyme polymorphism should be checked before starting azathioprine, and why?
3. Name a drug class used for IBS-C and one used for IBS-D that are NOT interchangeable.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

This lecture covers three separate drug territories: **Inflammatory Bowel Disease**, **Irritable Bowel Syndrome**, and **Acid-Peptic Disorders**. They don't overlap much, so each gets its own section below.

1 Inflammatory Bowel Disease (IBD)

Crohn's disease and ulcerative colitis. Clinical picture: bloody diarrhoea, mucus in stools, urgency, tenesmus, malaise, weight loss, fever.

Aminosalicylates (5-ASA drugs)

Drug	Key point
Sulphasalazine	Prodrug = sulphapyridine + 5-ASA. Colonic bacteria cleave it; the released 5-ASA acts locally. Sulphapyridine gets absorbed and causes the side effects: nausea, vomiting, headache, bone marrow suppression, rarely oligospermia. Allergic effects: rash, fever, hepatitis, pancreatitis, pneumonitis.
Mesalamine	Pure 5-ASA. Well absorbed in the upper GIT, so needs delayed-release capsules or pH-dependent tablets to reach the colon (or given as suppository/enema).
Olsalazine	Two 5-ASA molecules joined by an azo bond, cleaved by colonic bacteria. Diarrhoea is common with this one.
Balsalazide	Prodrug releasing 5-ASA after colonic hydrolysis of a diazo linkage.

Shared MOA: scavenges free radicals, inhibits COX and LOX (↓ prostaglandins/leukotrienes), ↓ neutrophil chemotaxis and superoxide generation via ↓ IL-1/TNF- α . Mesalamine/olsalazine/balsalazide all have fewer side effects than sulphasalazine. Mainly effective for **mild-moderate ulcerative colitis**.

Mnemonic — “*SOB needs colonic bacteria*”: **S**ulphasalazine, **O**lsalazine, **B**alsalazide all rely on colonic bacteria to release 5-ASA. Mesalamine doesn't — it needs a special formulation instead.

Glucocorticoids

Drugs	Prednisolone (oral), methylprednisolone (oral/parenteral), hydrocortisone (enema/suppository — distal colitis), budesonide (oral, high first-pass → fewer systemic effects)
Mechanism	Bind cytosolic glucocorticoid receptors → nuclear translocation → suppress transcription of pro-inflammatory genes (IL-2, TNF- α , IFN- γ)
Use	Short-term, to induce remission in moderate-severe disease — never for long-term maintenance
Side effects	HPA-axis suppression, osteoporosis, peptic ulcer, infections, hyperglycaemia, hypertension, cushingoid features

Immunosuppressants

Drug	Key point
Azathioprine / 6-mercaptopurine (thiopurines)	MOA: converted to 6-thioguanine nucleotides → inhibit purine synthesis → block DNA/RNA synthesis in lymphocytes. Delayed onset: 8–12 weeks. Metabolised by TPMT — polymorphisms affect toxicity. SE: myelosuppression, hepatotoxicity, pancreatitis, ↑ lymphoma risk.
Methotrexate	Oral/IM/SC, weekly. Inhibits dihydrofolate reductase → ↑ adenosine → ↓ inflammation. More effective in Crohn's than UC . SE: bone marrow suppression, hepatotoxicity, pulmonary fibrosis.

Mnemonic — check **TPMT before AZA/6-MP**: low activity means the drug piles up and myelosuppression follows.

Biological response modifiers & JAK inhibitors

Class	Drugs	MOA	Side effects
Anti-TNF	Infliximab, adalimumab, certolizumab, golimumab	Bind TNF- α , prevent TNFR interaction $\rightarrow$ block pro-inflammatory signalling	Infusion reactions, infections (esp. TB, Hep B reactivation), demyelination, lymphoma
Anti-integrin	Vedolizumab ($\alpha 4\beta 7$ antibody)	Blocks lymphocyte trafficking to the gut	Safer systemic profile; rare PML risk (lower than natalizumab)
Anti-IL-12/23	Ustekinumab	Binds the p40 subunit shared by IL-12/23 $\rightarrow$ inhibits Th1/Th17 pathways	Infections, rarely malignancy
JAK inhibitors (oral)	Tofacitinib (UC only), upadacitinib (UC + Crohn's)	Inhibit JAK1/3 $\rightarrow$ block STAT signalling $\rightarrow$ $\downarrow$ cytokine-driven inflammation	Infections (esp. herpes zoster), hyperlipidaemia, cytopenias, thrombosis

Used in severe Crohn's / refractory UC. Main drawbacks of biologics generally: cost, infection risk.

Antibiotics, probiotics & diet

Antibiotics used in IBD: metronidazole, ciprofloxacin, clarithromycin.

Probiotics (Lactobacillus, Streptococcus thermophilus, Bifidobacterium, Saccharomyces boulardii) are used as adjunctive therapy in UC/pouchitis, and more broadly for antibiotic-associated diarrhoea, infectious/traveller's diarrhoea, IBS, lactose intolerance, and as an H. pylori adjuvant. Use with caution in immunocompromised, critically ill, or central-line patients (sepsis risk).

Crohn's flare — CDED diet

Cut out	Wheat/gluten grains, dairy, animal fats, additives/ultra-processed foods, red meat
Encourage	Whole foods, fruits, vegetables, rice, potatoes, lean meats and eggs

2 Irritable Bowel Syndrome (IBS)

IBS-C (constipation-predominant)

Drug	Mechanism / note
Linaclotide	Guanylate cyclase-C agonist $\rightarrow$ $\uparrow$ cGMP $\rightarrow$ $\uparrow$ $\text{Cl}^-/\text{HCO}_3^-$ secretion $\rightarrow$ faster transit; also reduces abdominal pain
Lubiprostone	$\uparrow$ intestinal fluid secretion; approved for women (trials underenrolled men)
Plecanatide	Same class as linaclotide — increases intestinal fluid, eases pain
Tenapanor	Blocks sodium absorption $\rightarrow$ $\uparrow$ water in the intestine $\rightarrow$ softer stool, faster transit
Tegaserod	Reintroduced with restrictions — women <65 without cardiovascular risk only

IBS-D (diarrhoea-predominant)

Drug	Mechanism / note
Rifaximin	Gut-selective antibiotic; $\downarrow$ bacterial overgrowth; 2-week course can give lasting relief
Eluxadoline	μ/κ opioid agonist + δ antagonist on enteric neurons $\rightarrow$ $\downarrow$ motility and secretion. Low oral bioavailability $\rightarrow$ limited central opioid effects
Alosetron	Slows colonic transit; reserved for severe IBS-D in women who failed other treatment, due to rare serious side effects
Loperamide	OTC antidiarrhoeal, slows contractions; controls diarrhoea but not proven for pain
Bile acid binders (e.g. cholestyramine)	Help when excess colonic bile acids drive the diarrhoea

Pain & bloating (either subtype)

Drug	Note
Antispasmodics (dicyclomine, hyoscyamine)	Relax intestinal smooth muscle — relieve cramping

Peppermint oil	Enteric-coated capsules act as an antispasmodic; good for pain and bloating
TCAs (e.g. amitriptyline)	Low dose — relieves pain, normalises motility; useful in IBS-D (constipating side effect works in its favour)
SSRIs (e.g. fluoxetine)	Useful when anxiety/depression coexist; motility effect may suit IBS-C
Gabapentin / pregabalin	May relieve severe abdominal pain and bloating

3 Acid-Peptic Disorders

Class	Drugs
Antacids	Aluminium hydroxide, magnesium hydroxide, calcium carbonate, sodium bicarbonate
H2 receptor blockers	Ranitidine (less used now — safety concerns), famotidine, nizatidine
Proton pump inhibitors	Omeprazole, pantoprazole, esomeprazole, lansoprazole
Mucosal protective agents	Sucralfate, bismuth subsalicylate, misoprostol (also a prostaglandin analogue)

Mnemonic — all four PPIs on this list end in “-prazole”.

P Must-remember pearls

- Steroids induce remission in IBD — they are never a maintenance drug.
- Thiopurines take 8–12 weeks to work; check TPMT status before starting to avoid severe myelosuppression.
- Every anti-TNF biologic carries a TB and hepatitis B reactivation risk — screen before starting.
- Sulphasalazine, olsalazine, and balsalazide all need colonic bacteria to release 5-ASA; mesalamine uses a special coating instead.
- IBS-C and IBS-D drug classes are not interchangeable — matching the drug to the bowel-habit subtype is the exam-relevant skill.

Q Test yourself

1. A UC patient on long-term oral steroids develops cushingoid features and hyperglycaemia. What is the underlying prescribing error?

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| A) Steroids were combined with 5-ASA | B) Steroids were used for maintenance instead of just induction |
| C) The dose was too low | D) Steroids were given rectally instead of orally |

2. A patient started on azathioprine develops severe myelosuppression. Which enzyme should have been checked before starting?

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| A) CYP2D6 | B) TPMT |
| C) G6PD | D) DPD |

3. A Crohn's patient with untreated latent TB is started on infliximab. What is the main risk?

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| A) Pulmonary fibrosis | B) TB reactivation |
| C) Hypertension | D) Oligospermia |

4. An IBS-D patient on loperamide still has abdominal pain despite better stool consistency. Why?

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| A) Loperamide is not absorbed | B) Loperamide treats diarrhoea but is not proven to relieve pain |
| C) The dose is always too low | D) Loperamide blocks 5-HT3 receptors |

5. A patient is started on misoprostol to prevent an NSAID-induced ulcer. What is its relevant mechanism here?

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| A) H2 receptor blockade | B) Proton pump inhibition |
| C) Prostaglandin (PGE1) analogue — mucosal protection | D) Antacid neutralisation |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — B — glucocorticoids are for short-term induction of remission only; long-term use is what caused the cushingoid picture.
- 2 — B — TPMT. Low activity leads to drug accumulation and severe myelosuppression on azathioprine/6-MP.
- 3 — B — anti-TNF agents carry a recognised risk of reactivating TB (and hepatitis B), so screening comes first.
- 4 — B — loperamide slows intestinal contractions and helps diarrhoea, but it is not proven to relieve abdominal pain.
- 5 — C — misoprostol is a prostaglandin (PGE1) analogue, providing mucosal protection.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Jangi Shawkat Salai lecture slides, SGL 1 — covers IBD, IBS, and acid-peptic disorders.
Study alongside the lecture, not instead of it.